高阶复合偏函数与隐函数求导

课后作业：习题册 5.5

今天作业交到第 4 页

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电子科技大学

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内容目录

1 复合函数的高阶偏导数

2 隐函数定理

复合函数的高阶偏导数

复合函数的高阶偏导数有一定难度，下面以例题形式分析和研究.

例 10

设 $z = f\left(x^2y, \dfrac{y}{x}\right)$，$f$ 具有二阶连续偏导数，求 $\dfrac{\partial z}{\partial x}, \dfrac{\partial^2 z}{\partial x^2}, \dfrac{\partial^2 z}{\partial x \partial y}$.



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解：设 $u = x^2y$，$v = \dfrac{y}{x}$，则 $z = f(u, v)$，故

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} = 2xy\frac{\partial f}{\partial u} - \frac{y}{x^2}\frac{\partial f}{\partial v} = 2xy \cdot f_u(u, v) - \frac{y}{x^2} \cdot f_v(u, v).$$

其中 f_u 和 f_v 是关于 u, v 的函数，记 $\dfrac{\partial f}{\partial u} = f_u(u, v), \dfrac{\partial f}{\partial v} = f_v(u, v)$.

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解：设 $u=x^2y$，$v=\dfrac{y}{x}$，则 $z=f(u,v)$，故

$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial x}=2xy\frac{\partial f}{\partial u}-\frac{y}{x^2}\frac{\partial f}{\partial v}=2xy\cdot f_u(u,v)-\frac{y}{x^2}\cdot f_v(u,v).$$

其中 f_u 和 f_v 是关于 u,v 的函数，记 $\dfrac{\partial f}{\partial u}=f_u(u,v),\dfrac{\partial f}{\partial v}=f_v(u,v)$. 对于 f_u 和 f_v 使用链式法则得:

$$\begin{aligned}\frac{\partial^2 z}{\partial x^2}=&2y\frac{\partial f}{\partial u}+2xy\frac{\partial f_u}{\partial x}+\frac{2y}{x^3}\frac{\partial f}{\partial v}-\frac{y}{x^2}\frac{\partial f_v}{\partial x}\\=&2y\frac{\partial f}{\partial u}+2xy\left(\frac{\partial^2 f}{\partial u^2}\frac{\partial u}{\partial x}+\frac{\partial^2 f}{\partial u\partial v}\frac{\partial v}{\partial x}\right)+\frac{2y}{x^3}\frac{\partial f}{\partial v}-\frac{y}{x^2}\left(\frac{\partial^2 f}{\partial v\partial u}\frac{\partial u}{\partial x}+\frac{\partial^2 f}{\partial v^2}\frac{\partial v}{\partial x}\right).\end{aligned}$$

例 10 解答

因为 f 具有二阶连续偏导数，所以 $\dfrac{\partial^2 f}{\partial u \partial v} = \dfrac{\partial^2 f}{\partial v \partial u}$. 故

$$\frac{\partial^2 z}{\partial x^2} = 4x^2y^2\frac{\partial^2 f}{\partial u^2} - \frac{4y^2}{x}\frac{\partial^2 f}{\partial u \partial v} + \frac{y^2}{x^4}\frac{\partial^2 f}{\partial v^2} + 2y\frac{\partial f}{\partial u} + \frac{2y}{x^3}\frac{\partial f}{\partial v}.$$

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$$\begin{aligned}
\frac{\partial^2 z}{\partial x \partial y} =& 2x\frac{\partial f}{\partial u} + 2xy\frac{\partial f_u}{\partial y} - \frac{1}{x^2}\frac{\partial f}{\partial v} - \frac{y}{x^2}\frac{\partial f_v}{\partial y} \\
=& 2x\frac{\partial f}{\partial u} + 2xy\left(\frac{\partial^2 f}{\partial u^2}\frac{\partial u}{\partial y} + \frac{\partial^2 f}{\partial u \partial v}\frac{\partial v}{\partial y}\right) \\
& - \frac{1}{x^2}\frac{\partial f}{\partial v} - \frac{y}{x^2}\left(\frac{\partial^2 f}{\partial v \partial u}\frac{\partial u}{\partial y} + \frac{\partial^2 f}{\partial v^2}\frac{\partial v}{\partial y}\right) \\
=& 2x^3y\frac{\partial^2 f}{\partial u^2} + y\frac{\partial^2 f}{\partial u \partial v} - \frac{y}{x^3}\frac{\partial^2 f}{\partial v^2} + 2x\frac{\partial f}{\partial u} - \frac{1}{x^2}\frac{\partial f}{\partial v}.
\end{aligned}$$

高阶导数的记号

为了数学简便，引进符号:

- $f_1 = \dfrac{\partial f}{\partial u} = f_u$，表示函数 f 对第一个中间变量 u 求偏导数;



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- $f_{12} = \dfrac{\partial^2 f}{\partial u \partial v} = f_{uv}$，表示函数 f 先对 u 再对 v 求偏导数，以此类推.



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则例 10 结果可记为:

$$
\begin{aligned}
\frac{\partial z}{\partial x} &= 2xy\frac{\partial f}{\partial u} - \frac{y}{x^2}\frac{\partial f}{\partial v} = 2xyf_1 - \frac{y}{x^2}f_2, \\
\frac{\partial^2 z}{\partial x^2} &= 4x^2y^2\frac{\partial^2 f}{\partial u^2} - \frac{4y^2}{x}\frac{\partial^2 f}{\partial u \partial v} + \frac{y^2}{x^4}\frac{\partial^2 f}{\partial v^2} + 2y\frac{\partial f}{\partial u} + \frac{2y}{x^3}\frac{\partial f}{\partial v} \\
&= 4x^2y^2f_{11} - \frac{4y^2}{x}f_{12} + \frac{y^2}{x^4}f_{22} + 2yf_1 + \frac{2y}{x^3}f_2, \\
\frac{\partial^2 z}{\partial x \partial y} &= 2x^3y\frac{\partial^2 f}{\partial u^2} + y\frac{\partial^2 f}{\partial u \partial v} - \frac{y}{x^3}\frac{\partial^2 f}{\partial v^2} + 2x\frac{\partial f}{\partial u} - \frac{1}{x^2}\frac{\partial f}{\partial v} \\
&= 2x^3yf_{11} + yf_{12} - \frac{y}{x^3}f_{22} + 2xf_1 - \frac{1}{x^2}f_2.
\end{aligned}
$$

复合高阶导数相关例题

例 11

设 $w = f(x+y+z, xyz)$，f 具有二阶连续偏导数，求 $\dfrac{\partial w}{\partial x}$ 和 $\dfrac{\partial^2 w}{\partial x \partial z}$.

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设 $w = f(x+y+z, xyz)$，f 具有二阶连续偏导数，求 $\dfrac{\partial w}{\partial x}$ 和 $\dfrac{\partial^2 w}{\partial x \partial z}$.

解：设 $u = x+y+z, v = xyz$，则 $w = f(u,v)$，故

$$\frac{\partial w}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} = f_1 + yzf_2,$$

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$$\frac{\partial w}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} = f_1 + yzf_2,$$

又因为 $f_1 = f_1(x+y+z, xyz), f_2 = f_2(x+y+z, xyz)$，所以

$$\frac{\partial^2 w}{\partial x \partial z} = \frac{\partial (f_1 + yzf_2)}{\partial z} = \frac{\partial f_1}{\partial z} + yf_2 + yz\frac{\partial f_2}{\partial z}.$$

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例 11

设 $w=f(x+y+z,xyz)$，f 具有二阶连续偏导数，求 $\dfrac{\partial w}{\partial x}$ 和 $\dfrac{\partial^2 w}{\partial x\partial z}$.

解：设 $u=x+y+z, v=xyz$，则 $w=f(u,v)$，故

$$\frac{\partial w}{\partial x}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial x}=f_1+yzf_2,$$

又因为 $f_1=f_1(x+y+z,xyz), f_2=f_2(x+y+z,xyz)$，所以

$$\frac{\partial^2 w}{\partial x\partial z}=\frac{\partial\left(f_1+yzf_2\right)}{\partial z}=\frac{\partial f_1}{\partial z}+yf_2+yz\frac{\partial f_2}{\partial z}.$$

又

$$\frac{\partial f_1}{\partial z}=\frac{\partial f_1}{\partial u}\frac{\partial u}{\partial z}+\frac{\partial f_1}{\partial v}\frac{\partial v}{\partial z}=f_{11}+xyf_{12},\quad \frac{\partial f_2}{\partial z}=f_{12}+xyf_{22}.$$

故

$$\frac{\partial^2 w}{\partial x\partial z}=f_{11}+y(x+z)f_{12}+xy^2zf_{22}+yf_2.$$

全微分相关练习

练习 4

$z = f\left(xy, x^2 + y^2\right)$，$f$ 有二阶连续偏导数，则 $\dfrac{\partial^2 z}{\partial x \partial y} = (\quad)$

$(A).f_1 + xyf_{11} + \left(y^2 + x^2\right) f_{21} + 4xyf_{22}$

$(B).f_1 + 2xyf_{11} + \left(y^2 + x^2\right) f_{12} + 4xyf_{22}$

$(C).f_1 + xyf_{11} + \left(2y^2 + 2x^2\right) f_{12} + 4xyf_{22}$

$(D).f_1 + xyf_{11} + \left(2y^2 + 2x^2\right) f_{12} + xyf_{22}$

全微分相关练习

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$z = f\left(xy, x^2 + y^2\right)$，$f$ 有二阶连续偏导数，则 $\dfrac{\partial^2 z}{\partial x \partial y} = (\quad)$

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$(C).f_1 + xyf_{11} + \left(2y^2 + 2x^2\right) f_{12} + 4xyf_{22}$

$(D).f_1 + xyf_{11} + \left(2y^2 + 2x^2\right) f_{12} + xyf_{22}$

解：直接计算有：

$$\frac{\partial z}{\partial x} = yf_1 + 2xf_2 \quad \frac{\partial^2 z}{\partial x \partial y} = f_1 + y\frac{\partial f_1}{\partial y} + 2x\frac{\partial f_2}{\partial y}.$$

我们注意到 $f_1 = f_1\left(xy, x^2 + y^2\right), f_2 = f_2\left(xy, x^2 + y^2\right)$，则

$$\begin{aligned}\frac{\partial^2 z}{\partial x \partial y} &= f_1 + y\left(xf_{11} + 2yf_{12}\right) + 2x\left(xf_{21} + 2yf_{22}\right) \\ &= f_1 + xyf_{11} + \left(2y^2 + 2x^2\right) f_{12} + 4xyf_{22}.\end{aligned}$$

全微分相关练习

练习 5

若 $z = x^n f\left(\frac{y}{x^2}, x\right)$，$f(u, v)$ 有二阶连续偏导数，则 $\frac{\partial^2 z}{\partial x \partial y} = (\quad)$

$(A).(n-2)x^{n-3}f_1 - 2x^{n-5}yf_{11} + x^{n-2}f_{21};$

$(B).(n-2)x^{n-2}f_1 - 2x^{n-3}yf_{11} + x^{n-1}f_{21};$

$(C).nx^{n-3}f_1 - 2x^{n-5}yf_{11} + x^{n-2}f_{21};$

$(D).(n-2)x^{n-3}f_1 - 2x^{n-4}yf_{11} + x^{n-2}f_{21}.$

练习 5 解答

解：直接计算有

$$\begin{aligned}\frac{\partial z}{\partial x} &= nx^{n-1}f\left(\frac{y}{x^2}, x\right) + x^n\left(-\frac{2y}{x^3}f_1 + f_2\right)\\ &= nx^{n-1}f - 2x^{n-3}yf_1 + x^n f_2,\end{aligned}$$

练习 5 解答

解: 直接计算有

$$\begin{aligned}
\frac{\partial z}{\partial x} &= nx^{n-1} f\left(\frac{y}{x^2}, x\right) + x^n \left(-\frac{2y}{x^3} f_1 + f_2\right) \\
&= nx^{n-1} f - 2x^{n-3} y f_1 + x^n f_2, \\
\frac{\partial^2 z}{\partial x \partial y} &= nx^{n-1} \frac{\partial f}{\partial y} - 2x^{n-3} f_1 - 2x^{n-3} y \frac{\partial f_1}{\partial y} + x^n \frac{\partial f_2}{\partial y}, \\
&= nx^{n-1} \cdot \frac{1}{x^2} f_1 - 2x^{n-3} f_1 - 2x^{n-3} y \cdot \frac{1}{x^2} f_{11} + x^n \cdot \frac{1}{x^2} f_{21} \\
&= nx^{n-3} f_1 - 2x^{n-3} f_1 - 2x^{n-5} y f_{11} + x^{n-2} f_{21} \\
&= (n-2) x^{n-3} f_1 - 2x^{n-5} y f_{11} + x^{n-2} f_{21}.
\end{aligned}$$

故选择 (A).

内容目录 ☞

1 复合函数的高阶偏导数

2 隐函数定理
- 一个方程的情形
- 方程组情形

隐函数定理

方程 $F(x,y)=0$ 确定隐函数的情形:



隐函数定理

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定理 1（隐函数存在定理）

设函数 $F(x,y)$ 在点 $P(x_0,y_0)$ 的某邻域内有连续偏导数，且

$$F(x_0,y_0)=0,\quad F_y(x_0,y_0)\neq 0.$$

则方程 $F(x,y)=0$ 在点 $P(x_0,y_0)$ 的某邻域内能唯一确定一个单值连续且有连续导数的函数 $y=f(x)$，使得 $F[x,f(x)]=0$ 且 $y_0=f(x_0)$，并有:

$$\frac{dy}{dx}=-\frac{F_x}{F_y}.$$

定理 1 的中求导公式的推导

将定理 1 中所确定的函数 $y = f(x)$ 代入方程 $F(x, y) = 0$，则有恒等式

$$F[x, f(x)] \equiv 0.$$

其中 x 是最终自变量.



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$$F[x, f(x)] \equiv 0.$$

其中 x 是最终自变量. 由全导数的求导公式有

$$F_x + F_y \frac{dy}{dx} = 0.$$

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$$F_x + F_y \frac{dy}{dx} = 0.$$

因为 F_y 连续，并且 $F_y(x_0, y_0) \neq 0$，所以由连续函数的保号性知存在 (x_0, y_0) 的某一邻域，在该邻域内 $F_y \neq 0$. 于是

$$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

隐函数求导相关例题

例 1

已知 $\ln\sqrt{x^2+y^2}=\arctan\dfrac{y}{x}$，求 $\dfrac{dy}{dx}$.

隐函数求导相关例题

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已知 $\ln\sqrt{x^2+y^2}=\arctan\dfrac{y}{x}$，求 $\dfrac{dy}{dx}$.

解：令 $F(x,y)=\ln\sqrt{x^2+y^2}-\arctan\dfrac{y}{x}$，则 $\dfrac{dy}{dx}=-\dfrac{F_x}{F_y}$.



隐函数求导相关例题

例 1

已知 $\ln\sqrt{x^2+y^2}=\arctan\dfrac{y}{x}$，求 $\dfrac{dy}{dx}$.

解：令 $F(x,y)=\ln\sqrt{x^2+y^2}-\arctan\dfrac{y}{x}$，则 $\dfrac{dy}{dx}=-\dfrac{F_x}{F_y}$. 又

$$
\begin{aligned}
F_x &= \frac{1}{\sqrt{x^2+y^2}}\cdot\frac{2x}{2\sqrt{x^2+y^2}}-\frac{1}{1+\left(\frac{y}{x}\right)^2}\cdot\left(-\frac{y}{x^2}\right)\\
&= \frac{x}{x^2+y^2}+\frac{y}{x^2+y^2}=\frac{x+y}{x^2+y^2},\\
F_y &= \frac{1}{\sqrt{x^2+y^2}}\cdot\frac{2y}{2\sqrt{x^2+y^2}}-\frac{1}{1+\left(\frac{y}{x}\right)^2}\cdot\frac{1}{x}\\
&= \frac{y}{x^2+y^2}-\frac{x}{x^2+y^2}=\frac{y-x}{x^2+y^2}.
\end{aligned}
$$

隐函数求导相关例题

例 1

已知 $\ln\sqrt{x^2+y^2}=\arctan\dfrac{y}{x}$，求 $\dfrac{dy}{dx}$.

解：令 $F(x,y)=\ln\sqrt{x^2+y^2}-\arctan\dfrac{y}{x}$，则 $\dfrac{dy}{dx}=-\dfrac{F_x}{F_y}$. 又

$$
\begin{aligned}
F_x &= \frac{1}{\sqrt{x^2+y^2}}\cdot\frac{2x}{2\sqrt{x^2+y^2}}-\frac{1}{1+\left(\frac{y}{x}\right)^2}\cdot\left(-\frac{y}{x^2}\right)\\
&= \frac{x}{x^2+y^2}+\frac{y}{x^2+y^2}=\frac{x+y}{x^2+y^2},\\
F_y &= \frac{1}{\sqrt{x^2+y^2}}\cdot\frac{2y}{2\sqrt{x^2+y^2}}-\frac{1}{1+\left(\frac{y}{x}\right)^2}\cdot\frac{1}{x}\\
&= \frac{y}{x^2+y^2}-\frac{x}{x^2+y^2}=\frac{y-x}{x^2+y^2}.
\end{aligned}
$$

因此，$\dfrac{dy}{dx}=-\dfrac{F_x}{F_y}=-\dfrac{x+y}{y-x}=\dfrac{x+y}{x-y}$.

例 1 解法二

解法二： 两边同时对 x 求导得:

$$\frac{1}{\sqrt{x^2+y^2}}\cdot\frac{1}{2\sqrt{x^2+y^2}}\cdot\left(2x+2yy'\right)=\frac{1}{1+\left(\frac{y}{x}\right)^2}\cdot\frac{xy'-y}{x^2}$$

从而

$$\frac{x+yy'}{x^2+y^2}=\frac{xy'-y}{x^2+y^2}$$

因此，$\dfrac{dy}{dx}=y'=\dfrac{x+y}{x-y}$.

例 1 解法二

解法二： 两边同时对 x 求导得:

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从而

$$\frac{x+yy'}{x^2+y^2}=\frac{xy'-y}{x^2+y^2}$$

因此，$\dfrac{dy}{dx}=y'=\dfrac{x+y}{x-y}$.

问题:

公式法与直接求导法各自的特点是什么？

隐函数定理 II

方程 $F(x,y,z)=0$ 确定隐函数的情形:

定理 2（隐函数存在定理 II）

设函数 $F(x,y,z)$ 在点 $P(x_0,y_0,z_0)$ 的某一邻域内有连续的偏导数，并且满足

$$F(x_0,y_0,z_0)=0,\quad F_z(x_0,y_0,z_0)\neq 0,$$

则方程 $F(x,y,z)=0$ 在点 $P(x_0,y_0,z_0)$ 的某一邻域内恒能唯一确定一个单值、连续且有连续偏导数的函数 $z=f(x,y)$，它满足条件 $z_0=f(x_0,y_0)$，并有:

$$\frac{\partial z}{\partial x}=-\frac{F_x}{F_z},\quad \frac{\partial z}{\partial y}=-\frac{F_y}{F_z}.$$

定理 2 的中求导公式的推导

将定理 2 中所确定的函数 $z = f(x, y)$ 代入方程 $F(x, y, z) = 0$，则有恒等式

$$F[x, y, f(x, y)] \equiv 0.$$

其中 x, y 是自变量. 由多元复合函数的链式法则，求偏导数有

$$F_x + F_z \frac{\partial z}{\partial x} = 0, \quad F_y + F_z \frac{\partial z}{\partial y} = 0.$$

定理 2 的中求导公式的推导

将定理 2 中所确定的函数 $z = f(x, y)$ 代入方程 $F(x, y, z) = 0$，则有恒等式

$$F[x, y, f(x, y)] \equiv 0.$$

其中 x, y 是自变量. 由多元复合函数的链式法则，求偏导数有

$$F_x + F_z \frac{\partial z}{\partial x} = 0, \quad F_y + F_z \frac{\partial z}{\partial y} = 0.$$

因为 F_y 连续，并且 $F_z(x_0, y_0, z_0) \neq 0$，所以由连续函数的保号性知存在 (x_0, y_0, z_0) 的某一邻域，在该邻域内 $F_z \neq 0$. 于是有

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}.$$

隐函数求导相关例题

例 2

设 $x^2+y^2+z^2-4z=0$，求 $\dfrac{\partial^2 z}{\partial x^2}$.

隐函数求导相关例题

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解：令 $F(x,y,z)=x^2+y^2+z^2-4z$，则 $F_x=2x, F_z=2z-4$. 故

$$\frac{\partial z}{\partial x}=-\frac{F_x}{F_z}=\frac{x}{2-z}.$$

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进而

$$\frac{\partial^2 z}{\partial x^2}=\frac{(2-z)+x\frac{\partial z}{\partial x}}{(2-z)^2}=\frac{(2-z)+x\cdot\frac{x}{2-z}}{(2-z)^2}=\frac{(2-z)^2+x^2}{(2-z)^3}.$$

隐函数求导相关例题

例 3

设 $z = f(x+y+z, xyz)$，求 $\dfrac{\partial z}{\partial x}, \dfrac{\partial x}{\partial y}, \dfrac{\partial y}{\partial z}$.

隐函数求导相关例题

例 3

设 $z = f(x+y+z, xyz)$，求 $\dfrac{\partial z}{\partial x}, \dfrac{\partial x}{\partial y}, \dfrac{\partial y}{\partial z}$.

解一 (公式法)：令 $F(x,y,z) = f(x+y+z, xyz) - z$，则

$$F_x = f_1 + yzf_2, \quad F_y = f_1 + xzf_2, \quad F_z = f_1 + xyf_2 - 1.$$

隐函数求导相关例题

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从而代入定理 2 中公式有：

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{f_1 + yzf_2}{f_1 + xyf_2 - 1},$$

$$\frac{\partial x}{\partial y} = -\frac{F_y}{F_x} = -\frac{f_1 + xzf_2}{f_1 + yzf_2},$$

$$\frac{\partial y}{\partial z} = -\frac{F_z}{F_y} = -\frac{f_1 + xyf_2 - 1}{f_1 + xzf_2}.$$

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例 3

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$$\frac{\partial y}{\partial z} = -\frac{F_z}{F_y} = -\frac{f_1 + xyf_2 - 1}{f_1 + xzf_2}.$$

公式法特点：

1. 化为 $F(x,y,z) = 0$，并把 F 看作自变量为 x, y, z 的三元函数；
2. 求出 F_x, F_y, F_z，再代入隐函数偏导数公式计算.

例 3 解答续

解法二 (直接求导法):

- 求 $\dfrac{\partial z}{\partial x}$, 两边对 x 求偏导 (z 是因变量, x 是自变量, 即 $z = z(x, y)$):

$$\frac{\partial z}{\partial x} = \left(1 + \frac{\partial z}{\partial x}\right) f_1 + f\left(yz + xy\frac{\partial z}{\partial x}\right) f_2 \Rightarrow \frac{\partial z}{\partial x} = \frac{f_1 + yzf_2}{1 - f_1 - xyf_2}.$$

例 3 解答续

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- 求 $\dfrac{\partial x}{\partial y}$, 两边对 y 求偏导 (x 是因变量, y 是自变量, 即 $x = x(y, z)$):

$$0 = \left(\frac{\partial x}{\partial y} + 1\right) f_1 + \left(yz\frac{\partial x}{\partial y} + xz\right) f_2 \Rightarrow \frac{\partial x}{\partial y} = -\frac{f_1 + xzf_2}{f_1 + yzf_2}$$



例 3 解答续

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- 求 $\dfrac{\partial y}{\partial z}$, 两边对 z 求偏导 (y 是因变量, z 是自变量, 即 $y = y(x, z)$):

$$1 = \left(\frac{\partial y}{\partial z} + 1\right) f_1 + \left(xz\frac{\partial y}{\partial z} + xy\right) f_2 \Rightarrow \frac{\partial y}{\partial z} = \frac{1 - f_1 - xyf_2}{f_1 + xzf_2}.$$

例 3 解答续

解法二 (直接求导法):

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直接求导法要点:

1. 分清因变量与自变量;
2. 按求导对象结构, 选择合适法则计算.

隐函数求导相关例题

例 4

设方程 $x+y-e^z=z$ 确定 $z=z(x,y)$，求 $\left.\dfrac{\partial^2 z}{\partial x^2}\right|_{(1,0)}$.

隐函数求导相关例题

例 4

设方程 $x+y-e^z=z$ 确定 $z=z(x,y)$，求 $\left.\dfrac{\partial^2 z}{\partial x^2}\right|_{(1,0)}$.

解：将方程两端分别关于 x 求偏导数，得：

$$1-e^z\frac{\partial z}{\partial x}=\frac{\partial z}{\partial x},\quad \text{即：}\ \frac{\partial z}{\partial x}=\frac{1}{1+e^z}.$$

再将上式两端分别对 x 求偏导数，得：

$$\frac{\partial^2 z}{\partial x^2}=-\frac{e^z\cdot\frac{\partial z}{\partial x}}{(1+e^z)^2}=-\frac{e^z}{(1+e^z)^3}.$$

隐函数求导相关例题

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再将上式两端分别对 x 求偏导数，得：

$$\frac{\partial^2 z}{\partial x^2}=-\frac{e^z\cdot\frac{\partial z}{\partial x}}{(1+e^z)^2}=-\frac{e^z}{(1+e^z)^3}.$$

将 $x=1,y=0$ 代入原方程 $x+y-e^z=z$ 有 $z=0$. 从而

$$\left.\frac{\partial^2 z}{\partial x^2}\right|_{(1,0)}=-\frac{e^0}{(1+e^0)^3}=-\frac{1}{8}.$$

隐函数求导相关练习

练习 1

设函数 $z(x,y)$ 由方程 $F\left(x+\frac{z}{y}, y+\frac{z}{x}\right)=0$ 所确定，则

$$x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=(\quad)$$

A. $z-xy$　　B. $z+xy$　　C. $xy-z$　　D. $xy+2z$

练习 2

设 $u=f(x,y,z), \varphi\left(x^2, e^y, z\right)=0, y=\sin x$，且 f, φ 可微，求 $\frac{du}{dx}$.

隐函数求导相关练习

练习 3

设方程 $x^2+y^2+z^2=yf\left(\frac{z}{y}\right)$ 确定 $z=z(x,y)$，其中 $f(u)$ 对 u 可微，求 $\frac{\partial z}{\partial x},\frac{\partial z}{\partial y}$.

练习 4

设函数 $f(u,v)$ 可微，$z=z(x,y)$ 由方程

$$(x+1)z-y^2=x^2f(x-z,y)$$

确定，求 dz.

隐函数求导相关练习

练习 3

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确定，求 dz.

- 有哪些方法可以解出上述题目？
- 这些方法的特点是什么？
- 你能总结出各种方法的最佳使用场景吗？

隐函数定理 III

方程组 $\begin{cases} F(x,y,u,v)=0 \\ G(x,y,u,v)=0 \end{cases}$ 确定隐函数的情形:

隐函数定理 III

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隐函数定理 3:

设 $F(x,y,u,v), G(x,y,u,v)$ 在点 $P(x_0,y_0,u_0,v_0)$ 的某邻域内有对各个变量的连续偏导数, 且

$$F(x_0,y_0,u_0,v_0)=0, \quad G(x_0,y_0,u_0,v_0)=0,$$

且偏导数所组成的函数行列式 (雅可比行列式)

$$J=\frac{\partial(F,G)}{\partial(u,v)}=\begin{vmatrix} \frac{\partial F}{\partial u} & \frac{\partial F}{\partial v} \\ \frac{\partial G}{\partial u} & \frac{\partial G}{\partial v} \end{vmatrix}$$

在点 P 处不等于零.

隐函数定理 III

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在点 P 处不等于零. 则方程组 $\begin{cases} F(x,y,u,v)=0 \\ G(x,y,u,v)=0 \end{cases}$ 在点 P 的某一邻域内可唯一确定一组单值连续且有连续偏导数的函数

$$u=u(x,y), \quad v=v(x,y).$$

隐函数定理 III 续

它们满足条件 $u_0 = u(x_0, y_0), v_0 = v(x_0, y_0)$，以及

$$F[x, y, u(x, y), v(x, y)] = 0, \quad G[x, y, u(x, y), v(x, y)] = 0.$$

并且 u, v 都具有关于 x, y 的连续偏导数，这些偏导数为：

$$\frac{\partial u}{\partial x} = -\frac{1}{J}\frac{\partial(F, G)}{\partial(x, v)} = -\frac{\begin{vmatrix} F_x & F_v \\ G_x & G_v \end{vmatrix}}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}}, \quad \frac{\partial v}{\partial x} = -\frac{1}{J}\frac{\partial(F, G)}{\partial(u, x)} = -\frac{\begin{vmatrix} F_u & F_x \\ G_u & G_x \end{vmatrix}}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}},$$

$$\frac{\partial u}{\partial y} = -\frac{1}{J}\frac{\partial(F, G)}{\partial(y, v)} = -\frac{\begin{vmatrix} F_y & F_v \\ G_y & G_v \end{vmatrix}}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}}, \quad \frac{\partial v}{\partial y} = -\frac{1}{J}\frac{\partial(F, G)}{\partial(u, y)} = -\frac{\begin{vmatrix} F_u & F_y \\ G_u & G_y \end{vmatrix}}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}},$$

其中 $J = \frac{\partial(F,G)}{\partial(u,v)} = \begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}$.

"Invert, always invert" —C. Jacobi